

The Dimension of Nonterminating Resampling Computations

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

July 19, 2026

Abstract

A randomized algorithm may terminate almost surely even though exceptional random tapes make it run forever. This paper studies the survival tail, the Kolmogorov complexity of one such tape, and the Hausdorff dimension of all of them. For matrix-commutative local repair, the main theorem bounds $\sum_w P[w]^s$ over surviving tape prefixes w , for every $s > 0$ and uniformly over deterministic nonanticipating selectors. At $s = 1$ it gives termination bounds; varying s gives dimension bounds and robustness to history-dependent weak randomness. For bounded-dependence k -SAT with clause-overlap graph D , conditional block min-entropy above $\log_2 \kappa_D$ gives exponential termination, where κ_D is the trace-growth rate. Tree and clique formulas make the stated dimension and one-step thresholds sharp in their respective regimes. Persistent recoloring gives a genuinely nonuniform application: a one-parameter matching family has the same complete running-time law throughout, while its nontermination dimension ranges from $1/2$ to arbitrarily near zero. A backward likelihood identity complements the setwise theorem with tail and coding bounds for each run.

1 Introduction

A randomized algorithm is deterministic once its random tape is fixed. Although the algorithm may terminate with probability one, some tapes may make it run forever. Let $\mathbb{P}$ denote the source law of the tape and put

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where τ is the number of repair steps before stopping. The tail $\mathbb{P}\{\tau \geq T\}$ measures how often the computation survives T steps. When the following extended limit exists, the *survival exponent* is

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This is the base-two form of the usual escape rate (Keller and Liverani, 2009).

The survival exponent does not describe the exceptional tapes themselves. For a computable process with a computable tape law P , $\mathcal{N}$ is effectively closed because termination, when it occurs, is determined by a finite prefix. Thus a P -null $\mathcal{N}$ contains no P -Martin-Löf-random tape, but it may still have positive or full Hausdorff dimension and contain tapes of maximal effective strong dimension. Effective dimension measures the algorithmic randomness of one tape; source-relative Hausdorff dimension measures the size of the whole exceptional set.

The Lovász local lemma has a different scope. It gives conditions under which all bad events can be avoided, and its algorithmic forms give termination or tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris,

2017). The present paper fixes the resampling computation and asks for its survival rate and for the complexity and geometry of its exceptional tapes. These are process-specific questions, rather than a stronger universal local-lemma criterion. The local lemma is valuable because it supplies efficiently checkable quantitative bounds without constructing the full assignment chain.

The motivation is pointwise. In statistical learning, the learned rule $\hat{h}(S)$ depends on the sample S , so a bound for a fixed rule cannot simply be evaluated at $\hat{h}(S)$; uniform and localized empirical-process methods control the data-dependent region in which it lies (Bartlett et al., 2005; Koltchinskii, 2006; Xu and Zeevi, 2025). A resampling computation creates an analogous dependence between a tape and the repair history generated from it: survival for T repairs restricts tape space to a nested family of live cylinders. Recent work studies complexity at one trained network or one index of a random field (Li and Xu, 2026; Xu, 2026b); here the pointwise object is one random tape. This is an analogy, not a reduction to a learning bound: it isolates the common question of how much complexity remains after data-dependent localization. The search for a smaller process state that preserves future source-power sums likewise parallels Bellman-sufficient representations (Xu, 2026a). Corollary 3.2 makes one such localization exact: the asymptotic complexity of an infinite k -SAT run depends only on the clauses repaired infinitely often.

For a computable full-support source $P = \pi^{\mathbb{N}}$, let $\omega_{1:n}$ denote the first n source symbols, and let

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}]$$

be their self-information. If $K(w)$ is prefix-free Kolmogorov complexity, the effective strong P -dimension of one tape is

$$\text{Dim}_{\text{eff},P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \quad (1)$$

This is the standard source-relative notion (Staiger, 1993; Athreya et al., 2007; Lutz, 2011). A value near one means that the tape retains nearly the full algorithmic information of the source; a smaller value means that continued execution has imposed additional structure. The source-relative Hausdorff dimension $\dim_H^P E$ of a tape set E is defined by assigning diameter $P[w]$ to the cylinder $[w]$ (Billingsley, 1965; Lutz, 2011). Section B gives the corresponding source tree when tape consumption is adaptive.

For the prefix-free family $\mathcal{L}_T$ of minimal tape prefixes that remain live for T repairs, define

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s. \quad (2)$$

This Rényi sum is the central object. At $s = 1$ it is the survival probability; its behavior across s controls Hausdorff dimension through the mass-distribution principle and Frostman’s lemma (Rényi, 1961; Falconer, 2014). It also turns a global min-entropy bound on a possibly dependent tape source into a running-time bound for the original algorithm.

1.1 Results

The main theorem is a source-power extension of matrix-commutative local repair. For each flaw f and power $s > 0$, the matrix $K_{f,s}$ sums the s th powers of action probabilities over state transitions. If the matrices for nonadjacent flaws commute, Theorem 2.3 bounds the generating series of $Z_T^\Lambda(s)$ by the Cartier–Foata/Shearer trace series, term by term and uniformly over every deterministic nonanticipating repair rule Λ . The exact-prefix proof works directly with the powered matrices and requires neither recovery of an action label from its endpoints nor an action-level swapping construction. At $s = 1$ the theorem gives termination bounds; varying s gives Hausdorff-dimension

bounds, effective-dimension bounds under computability, and direct guarantees for history-dependent weak sources.

For bounded-dependence k -SAT, conditional block min-entropy above $\log_2 \kappa_D$ gives an exponential tail, where κ_D is the trace-growth rate of the clause-overlap graph. Tree formulas with variable occurrence two asymptotically attain the maximum-degree dimension bound and its global source threshold, including the critical power. Clique formulas attain the graph-specific one-step threshold in the stated regime. The same analysis gives an end-to-end random-bit bound, applies directly to Santha–Vazirani tapes, localizes the complexity of an infinite run to the clauses repaired infinitely often, and extends beyond product state spaces to rainbow perfect matching. Persistent local recoloring supplies the nonuniform application absent from ordinary trace counting: its powered kernels are not scalar rescalings of their power-one counterparts. On a matching, the complete running-time law is fixed while the Hausdorff and maximal effective strong dimensions vary from $1/2$ to arbitrarily near zero; the same theorem gives selector-uniform bounds for overlapping repair on a path.

Complementing this setwise theorem, Theorem 4.1 reads one finite run backward and expresses its information surplus as an exact likelihood ratio. The same identity gives a tail bound and, under computability, a saving in the description length of every surviving tape prefix. The finite-state normalization and all proofs are deferred to the appendix.

1.2 Relation to earlier work

Reverse reconstruction is a basic entropy-compression device (Moser, 2009; Moser and Tardos, 2010). Messner and Thierauf reconstruct the uniform bits used by Moser’s satisfiability search and apply Kolmogorov complexity (Messner and Thierauf, 2012); Fialho, de Lima, and Procacci prove that the Moser–Tardos and entropy-compression bounds coincide through the subset-gas cluster expansion (Fialho et al., 2022). The complementary backward theorem here supplies an exact likelihood identity for general finite product sources and non-atomic variable-model events, permits an arbitrary comparison law Q_T , and identifies the slack as relative entropy.

Action-level commutativity for resampling oracles was developed by Kolmogorov (2018); the matrix formulation of Harris et al. (2025) is broader. At $s = 1$, their matrix analysis and the usual Cartier–Foata/Shearer enumeration control ordinary repair histories. Theorem 2.3 instead proves the full-prefix estimate for every source power at which the matrices commute, including powers for which their row sums exceed one, while allowing action collisions and selectors that use the complete action history. This is the step that yields weak-source and dimension bounds.

For extremal partial rejection, confluence and the exact probability-weighted trace series are known (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024); Bernoulli measures on infinite trace boundaries were characterized by Abbes and Mairesse (2015). The finite-state links among weighted growth, Hausdorff dimension, and Kolmogorov complexity also have a substantial history in symbolic dynamics and regular infinite languages (Staiger, 1993; Parry, 1964; Bowen, 1975; Mauldin and Williams, 1988; Simpson, 2015). The present use of this machinery is computational: source powers are inserted into local-repair matrices and trace series to analyze the exceptional tapes of a fixed algorithm.

The weak-source guarantee is instance-specific and narrower than general simulation from weak randomness (Chor and Goldreich, 1988; Doron et al., 2023). It leaves the selector and available actions unchanged and runs the original repair map without an extractor. Likewise, the powered theorem is an upper bound rather than a new universal LLL threshold. Exactness is established only for the clique, tree, and persistent-matching families, and general overlapping repair still lacks a compact exact representation.

Section 2 proves the powered matrix bound; Section 3 develops weak-source, localization, and sharpness consequences; and Section 4 gives the backward likelihood identity. The appendix contains the finite-state normalization, the consumed-tape model, and all proofs.

2 Source-power local repair

This section bounds the complete source-power sum of an overlapping repair process without constructing its full assignment chain. The input is a family of powered transition matrices and local scalar bounds on their growth. The conclusion is uniform over the order in which present flaws are selected.

Let Ω be a finite state space and let $F_\sigma \subseteq F$ be the flaws present at σ . Addressing $f \in F_\sigma$ chooses an action $a \in \mathcal{A}_f(\sigma)$ with probability $\rho_f(a \mid \sigma) > 0$ and moves to $\Phi_f(\sigma, a)$. The action sets are finite and satisfy

$$\sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma) = 1.$$

Distinct actions may lead to the same terminal state. A deterministic nonanticipating selector may use the complete observed past, including past action labels, but not future randomness.

Let $\sim$ be a symmetric relation on distinct flaws and define the inclusive neighborhood

$$\Gamma(f) = \{f\} \cup \{g \in F : g \sim f\}.$$

In local-repair applications, $\sim$ is also chosen to satisfy the potential-causality condition

$$F_{\Phi_f(\sigma, a)} \subseteq (F_\sigma \setminus \{f\}) \cup \Gamma(f). \quad (3)$$

This is Kolmogorov's convention after adding a self-loop at every flaw. The matrix theorem itself uses only powered commutation; potential causality explains why D is a local dependency graph. Write $\text{Ind}(S)$ for the subsets of S whose distinct elements are pairwise nonadjacent.

For $s > 0$, define the powered action kernel

$$K_{f,s}(\sigma, \tau) = \mathbf{1}\{f \in F_\sigma\} \sum_{\substack{a \in \mathcal{A}_f(\sigma) \\ \Phi_f(\sigma, a) = \tau}} \rho_f(a \mid \sigma)^s. \quad (4)$$

The kernel aggregates actions with the same endpoints, but for $s \neq 1$ its entries retain the powered probabilities of the individual actions and need not be determined by the ordinary transition matrix $K_{f,1}$. The process is *s-commutative* if

$$K_{f,s}K_{g,s} = K_{g,s}K_{f,s} \quad (f \neq g, f \not\sim g). \quad (5)$$

Two familiar conditions imply this matrix identity. First, suppose that for nonadjacent f, g , action-labelled paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

are in product-preserving bijection with paths

$$\sigma \xrightarrow{g,b'} \sigma'' \xrightarrow{f,a'} \tau,$$

so that

$$\rho_f(a \mid \sigma) \rho_g(b \mid \sigma') = \rho_g(b' \mid \sigma) \rho_f(a' \mid \sigma''). \quad (6)$$

Second, suppose that each enabled instance of flaw f has the same number M_f of equiprobable actions.

Lemma 2.1 (Two routes to powered commutation). *The product-preserving diamond property implies (5) for every $s > 0$. Alternatively, if each enabled f has exactly M_f actions of probability $1/M_f$, then*

$$K_{f,s} = M_f^{1-s} K_{f,1}. \quad (7)$$

Consequently ordinary matrix commutation at $s = 1$ implies (5) for every $s > 0$ in this fixed-arity case. Different action labels may lead to the same terminal state.

The proof is in Appendix B.

Ordinary matrix commutation at $s = 1$ need not survive powering. For example,

$$A_0 = \begin{pmatrix} 4/5 & 1/5 \\ 2/5 & 3/5 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 7/10 & 3/10 \\ 3/5 & 2/5 \end{pmatrix}$$

commute, whereas their entrywise square roots do not. Thus the full source-power family requires additional structure beyond ordinary matrix commutation; Lemma 2.1 gives two sufficient conditions.

Fix an initial law u on Ω . The initial source symbol is taken to be the initial state, and its powered mass vector is

$$b_s(\sigma) = u(\sigma)^s.$$

If instead an initial symbol α of probability $q(\alpha)$ produces the state $\iota(\alpha)$, replace this by

$$b_s(\sigma) = \sum_{\alpha: \iota(\alpha)=\sigma} q(\alpha)^s.$$

One later source symbol is one complete oracle action. Let $\mathcal{L}_T^\Lambda$ be the prefix-free family of minimal tape prefixes that complete T repairs under selector Λ , and set

$$Z_T^\Lambda(s) = \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s, \quad (8)$$

where $P_{\text{tape}}[w]$ includes the initial source symbol. With this convention, $Z_T^\Lambda(1) = \mathbb{P}\{\tau \geq T\}$ when τ is the number of completed repairs.

Write $f \simeq g$ when $f = g$ or $f \sim g$. For a flaw word $\ell = (f_1, \dots, f_T)$, its *full history DAG* has vertex set $[T]$ and an edge $i \rightarrow j$ exactly when $i < j$ and $f_i \simeq f_j$. For such a DAG H , put

$$K_{\emptyset,s} = I, \quad K_{H,s} = K_{L(x),s} K_{H-x,s},$$

where x is any source and $L(x)$ is its label. Under (5), this definition is independent of the chosen source.

Two full history DAGs are identified when they are related by a label-preserving directed-graph isomorphism.

Choose a strictly positive probability vector v_s on Ω and define

$$\lambda_f(s) = \max_{\tau \in \Omega} \frac{\sum_{\sigma \in \Omega} v_s(\sigma) K_{f,s}(\sigma, \tau)}{v_s(\tau)}, \quad (9)$$

$$\gamma_s = \max_{\sigma \in \Omega} \frac{b_s(\sigma)}{v_s(\sigma)}. \quad (10)$$

Let D be the graph on F with edges $f \sim g$ and let $\mathcal{M}(D)$ be the trace monoid in which nonadjacent flaws commute. For a trace h , write $N_f(h)$ for the multiplicity of f and $|h| = \sum_f N_f(h)$. Define

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{f \in F} \lambda_f(s)^{N_f(h)}, \quad (11)$$

$$P_D(\mathbf{z}) = \sum_{I \in \text{Ind}(F)} (-1)^{|I|} \prod_{f \in I} z_f. \quad (12)$$

The open Shearer region consists of the vectors $\mathbf{z} \geq 0$ for which $P_D(\mathbf{y}) > 0$ for every $0 \leq \mathbf{y} \leq \mathbf{z}$. The Cartier–Foata identity gives

$$\sum_{T \geq 0} a_T(s) z^T = \frac{1}{P_D(z\boldsymbol{\lambda}(s))} \quad (13)$$

as a formal power series, where $\boldsymbol{\lambda}(s) = (\lambda_f(s))_{f \in F}$.

For two formal series $F(z) = \sum_{T \geq 0} a_T z^T$ and $G(z) = \sum_{T \geq 0} b_T z^T$, write $F \preceq_{\text{term}} G$ when $a_T \leq b_T$ for every T .

At $s = 1$, [Harris et al. \(2025, Lemma 3.5\)](#) bound the probability that a prescribed history DAG appears during a run. Their induction allows the current repair to be absent from that DAG and uses substochasticity in this case. Here H records the entire next T -repair prefix, so the current repair must label a source of H . The omission case disappears, allowing the powered matrices to have row sums greater than one when $s < 1$.

Lemma 2.2 (Exact-prefix matrix bound). *Fix $s > 0$ for which (5) holds. For a full history DAG H with T vertices, let $\mathcal{L}_T^\Lambda(H)$ be the live prefixes whose first T repair labels induce H . Every deterministic nonanticipating selector satisfies*

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq b_s^\top K_{H,s} \mathbf{1}. \quad (14)$$

No injectivity of the action-to-state map is required.

For positive numbers $(\eta_f)_{f \in F}$, write $\eta(I) = \prod_{f \in I} \eta_f$ and put

$$\theta_s = \max_{f \in F} \frac{\lambda_f(s)}{\eta_f} \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I). \quad (15)$$

Theorem 2.3 (Powered matrix-commutative local lemma). *Fix $s > 0$ and suppose that the powered kernels satisfy (5). Then every deterministic nonanticipating selector Λ satisfies*

$$Z_T^\Lambda(s) \leq \gamma_s a_T(s) \quad (T \geq 0), \quad (16)$$

or, equivalently,

$$\sum_{T \geq 0} Z_T^\Lambda(s) z^T \preceq_{\text{term}} \frac{\gamma_s}{P_D(z\boldsymbol{\lambda}(s))}. \quad (17)$$

If $q\boldsymbol{\lambda}(s)$ lies in the open Shearer region for some $q > 1$, then

$$Z_T^\Lambda(s) \leq \frac{\gamma_s}{P_D(q\boldsymbol{\lambda}(s))} q^{-T}. \quad (18)$$

If instead $\theta_s < 1$, then

$$Z_T^\Lambda(s) \leq \gamma_s \left(\sum_{R \in \text{Ind}(F)} \eta(R) \right) \theta_s^T \quad (T \geq 0). \quad (19)$$

All quantities on the right are independent of the selector.

The term-by-term trace bound is the sharpest of these statements. The Shearer condition turns it into an exponential estimate, while the cluster expression is a convenient local sufficient condition. At $s = 1$ they recover established commutative-LLL bounds. The new use is to control the full source-power family, and hence weak randomness and exceptional-tape dimension, by the same matrices.

Whenever either strict condition in Theorem 2.3 holds, choose constants such that

$$Z_T^\Lambda(s) \leq C_s \vartheta_s^T, \quad C_s < \infty, \quad 0 < \vartheta_s < 1, \quad (20)$$

using the corresponding constants from (18) or (19). If the rate supplied by the theorem is zero, the same estimate remains valid after replacing it by any fixed number in $(0, 1)$.

Corollary 2.4 (Dimension and pointwise coding). *Assume (20) and $\rho_{\max} = \max_{\sigma, f, a} \rho_f(a \mid \sigma) < 1$. Then*

$$\dim_H^{P_{\text{tape}}} \mathcal{N} \leq s. \quad (21)$$

Suppose in addition that s , the process, the selector, and computable numbers $\widehat{C}_s \geq C_s$ and $0 < \widehat{\vartheta}_s < 1$ with $\widehat{\vartheta}_s \geq \vartheta_s$ are computable. Include fixed descriptions of these objects in instance. Every $w \in \mathcal{L}_T^\Lambda$ then satisfies

$$K(w \mid T, \text{instance}) \leq s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1), \quad (22)$$

and every $\omega \in \mathcal{N}$ has $\text{Dim}_{\text{eff}, P}(\omega) \leq s$ in this action-outcome source tree.

The metric in this corollary is tied to complete oracle actions. A binary random-bit implementation gives the same conclusion when it is related to this tree by a computable prefix-free sampler with bounded distortion; no invariance under arbitrary implementations is asserted.

Corollary 2.5 (Dependent repair choices). *Let ν be a possibly history-dependent law that uses only actions available at the corresponding reference history. If a constant $D < \infty$ satisfies*

$$\nu[w] \leq D P_{\text{tape}}[w]^s \quad (w \in \mathcal{L}_T^\Lambda, T \geq 0), \quad (23)$$

then

$$\mathbb{P}_\nu\{\tau \geq T\} \leq D C_s \vartheta_s^T. \quad (24)$$

When $0 < s \leq 1$ and the reference oracle is uniform on the same M_t actions available at step t , the explicit condition

$$\max_a \mathbb{P}_\nu\{A_t = a \mid \mathcal{H}_{t-1}\} \leq M_t^{-s}$$

for every feasible positive-probability history implies (23) with $D = 1$ when ν and P_{tape} also use the same initial law. The repair choices may be correlated across time, but may alter neither the selector nor the valid action sets. For a different initial law, or for $s > 1$, the corresponding initial likelihood ratio is absorbed into D when it is bounded.

3 Weak randomness, localization, and sharpness

For a variable-model event A , let $v(A)$ be its resampled coordinates and let $\pi_{v(A)}$ be their product law. Define its local source-power mass by

$$r_A(s) = \sum_{b \in A} \pi_{v(A)}(b)^s. \quad (25)$$

Here A is identified with its local configurations on $v(A)$. Taking $v_s(\sigma)$ proportional to $\pi(\sigma)^s$ in (9) gives the exact local charge

$$\lambda_A(s) = r_A(s). \quad (26)$$

Thus Theorem 2.3 can be checked from event scopes, local source masses, and the causality graph, without enumerating all assignments. Replacing the independent-set sum in (15) by $\prod_{B \in \Gamma(A)} (1 + \eta_B)$ gives a simpler polynomial-time sufficient test, though generally a weaker one. The independent-set refinement is the standard cluster-expansion form (Bissacot et al., 2011; Pegden, 2014); the use made here of the full source-power family is separate.

For a finite graph D , let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}. \quad (27)$$

Equivalently, $\kappa_D = R_D^{-1}$, where R_D is the first positive zero of $z \mapsto P_D(z\mathbf{1})$.

Theorem 3.1 (Graph-specific min-entropy threshold for k -SAT). *Let Φ be a k -CNF with clause-overlap graph D , start from a fixed assignment, and use any deterministic nonanticipating selector. Let P be the law of independent uniform k -bit redraws. If a possibly history-dependent redraw law ν satisfies, for some $D_0 < \infty$ and $s > 0$,*

$$\nu[w] \leq D_0 P[w]^s \quad (28)$$

for every finite live redraw word w , then, for every $\xi > \kappa_D$,

$$\mathbb{P}_\nu\{\tau \geq T\} \leq \frac{D_0 2^n}{P_D(\xi^{-1}\mathbf{1})} (\xi 2^{-ks})^T. \quad (29)$$

In particular, the tail is exponential whenever

$$s > \frac{\log_2 \kappa_D}{k}. \quad (30)$$

If $h > 0$ and each redraw has conditional min-entropy at least h , then (29) holds with $D_0 = 1$ and $ks = h$. Consequently $h > \log_2 \kappa_D$ implies exponential termination. For the uniform reference law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 \kappa_D}{k} \right\}. \quad (31)$$

If the selector is computable, the same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$.

If D has maximum degree at most $d \geq 2$, then

$$\kappa_D \leq c_d, \quad c_d = \frac{d^d}{(d-1)^{d-1}}. \quad (32)$$

Thus $h > \log_2 c_d$ is a selector-uniform sufficient condition, and the right side of (31) is at most $\min\{1, \log_2(c_d)/k\}$.

The one-step min-entropy hypothesis implies (28) by the chain rule. The converse can fail: a globally dominated source may move randomness between different times. The tree family below makes the global threshold and the dimension bound sharp, but does not by itself prove sharpness of the one-step conditional threshold $\log_2 c_d$.

For an infinite run ω , let

$$\text{Rec}(\omega) = \{A : A \text{ is repaired infinitely often on } \omega\}. \quad (33)$$

This set is nonempty because there are finitely many clauses.

Corollary 3.2 (Pointwise localization to recurrent clauses). *For a computable k -SAT repair process and every infinite run ω ,*

$$\text{Dim}_{\text{eff},P}(\omega) \leq \min \left\{ 1, \frac{\log_2 \kappa_{D[\text{Rec}(\omega)]}}{k} \right\}. \quad (34)$$

For each nonempty $R \subseteq F$, the same right side with $D[R]$ bounds the Hausdorff dimension of the tapes satisfying $\text{Rec}(\omega) = R$.

This is the precise localization statement behind the motivation in Section 1: clauses repaired only finitely often affect a finite prefix but not the asymptotic complexity of that tape.

For $d \geq 2$, define

$$A_d = 1 + \frac{d^d}{(d-1)^{d-1}}. \quad (35)$$

The graph-specific theorem is sharper but involves P_D . The following local relaxation has explicit constants and, relative to the worst-case max-degree bound c_d , loses at most $\log_2(A_d/c_d) \leq \log_2(5/4)$ bits per repair.

Theorem 3.3 (An explicit max-degree tail for k -SAT). *Let Φ be a k -CNF on n variables with m clauses, each containing k distinct variables and overlapping at most $d \geq 2$ other clauses. Start from an arbitrary assignment and use any deterministic nonanticipating rule to select a violated clause. Suppose that, at every feasible history, the next k -bit redraw B_t has worst-case conditional min-entropy at least h , in the pointwise sense*

$$\max_b \mathbb{P}\{B_t = b \mid \mathcal{H}_{t-1}\} \leq 2^{-h}. \quad (36)$$

If $\varepsilon = h - \log_2 A_d > 0$, then

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m 2^{-\varepsilon T}. \quad (37)$$

For independent uniform redraws, this becomes

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m \left(\frac{A_d}{2^k} \right)^T. \quad (38)$$

Writing P for the uniform independent repair-tape law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 A_d}{k} \right\}. \quad (39)$$

If the selector is computable, the same upper bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff},P}(\omega)$. The tail and Hausdorff-dimension conclusions are uniform over every deterministic nonanticipating selector and every history-dependent redraw law satisfying (36).

Corollary 3.4 (Fresh fair bits). *Before repair t , let the past determine an injective map*

$$E_t : \{0, 1\}^r \longrightarrow \{0, 1\}^k, \quad 1 \leq r \leq k,$$

then read r fresh independent fair bits U_t and redraw the selected clause to $E_t(U_t)$. If $\varepsilon_r = r - \log_2 A_d > 0$, then (37) holds with $\varepsilon = \varepsilon_r$, and

$$\mathbb{E}\tau \leq \left\lceil \frac{n + m \log_2(d/(d-1))}{\varepsilon_r} \right\rceil + \frac{1}{2^{\varepsilon_r} - 1}. \quad (40)$$

The expected number of fair bits is $r\mathbb{E}\tau$. In particular, if $A_d \leq 2^{k-1}$, taking $r = \lceil \log_2 A_d + 1 \rceil$ gives

$$\mathbb{E}[\text{fair bits}] = O((n + m/d) \log(d+1)). \quad (41)$$

The redraw map may change adaptively with the entire past, but must be fixed before U_t is read.

Messner and Thierauf's *Modified-Search* instead redraws a violated k -clause uniformly from its $2^k - 1$ satisfying local assignments and obtains the corresponding $c_d \leq 2^k - 1$ condition under their lopsided degree convention (Messner and Thierauf, 2012). Corollary 3.4 addresses a different resource question: the adaptive redraw may use only 2^r outcomes, and hence only $r = O(\log d)$ fresh fair bits, without requiring that the violated local assignment be excluded.

Corollary 3.5 (A Santha–Vazirani tape). *Let the bits (U_j) satisfy, for some $0 < \alpha \leq 1/2$,*

$$\alpha \leq \mathbb{P}\{U_j = 1 \mid U_1, \dots, U_{j-1}\} \leq 1 - \alpha$$

at every positive-probability history (Santha and Vazirani, 1986). Feed successive blocks of r bits through the adaptive injective maps in Corollary 3.4. If

$$\varepsilon_\alpha = r \log_2 \frac{1}{1 - \alpha} - \log_2 A_d > 0,$$

then (37) and (40) hold with ε_α in place of ε_r . Taking $r = k$ and E_t to be the identity runs ordinary clause resampling directly from the Santha–Vazirani tape, without an extractor.

Proposition 3.6 (A one-step min-entropy obstruction). *Let $d \geq 2$ and $k \geq 2$. If $d + 1 \leq 2^{k-1}$, there is a satisfiable k -CNF of dependency degree d whose repair scopes overlap without being equal. From a specified initial assignment, uniform independent k -bit redraws satisfy*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \frac{d+1}{2^k}, \quad (42)$$

$$\dim_H^P \mathcal{N} = \sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \frac{\log_2(d+1)}{k}. \quad (43)$$

There is also a fixed clause-wise redraw source of conditional block min-entropy $\log_2(d+1)$ on which every run is infinite. The dependency graph of this instance is K_{d+1} , for which $\kappa_{K_{d+1}} = d+1$. Thus Theorem 3.1 is sharp for this graph: strict inequality above $\log_2(d+1)$ gives exponential termination, whereas equality can sustain an infinite run. Among guarantees that use only the maximum degree d , the one-step sufficient and obstructing thresholds are separated by less than

$$\log_2 c_d - \log_2(d+1) < \log_2 e$$

bits. When $d+1 = 2^r$, the infinite-run source is an injective r -fair-bit redraw. This last source need not be Santha–Vazirani, so the proposition does not claim sharpness for Corollary 3.5.

Theorem 3.7 (Tree formulas approach the max-degree boundary). *Fix integers $2 \leq d \leq k$. For every height $\ell \geq 1$, there is a satisfiable k -CNF $\Phi_{d,k,\ell}$ whose clause-overlap graph is the full rooted $(d-1)$ -ary tree $T_{d,\ell}$ of height ℓ . Every variable occurs in at most two clauses, every clause overlaps at most d others, and every pair of adjacent bad events is disjoint. Draw the initial assignment uniformly, and then use independent uniform redraws; let P denote this sequential tape law. For any deterministic current-assignment selector,*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = 2^{-k} \kappa_{T_{d,\ell}}, \quad (44)$$

$$\dim_H^P \mathcal{N} = \frac{\log_2 \kappa_{T_{d,\ell}}}{k}. \quad (45)$$

If the selector is computable, the right side of (45) is also the maximal effective strong dimension. Moreover,

$$\kappa_{T_{d,\ell}} \uparrow \frac{d^d}{(d-1)^{d-1}} = c_d \quad (\ell \rightarrow \infty). \quad (46)$$

Thus the max-degree dimension bound in Theorem 3.1 is best possible in the supremum over finite instances, already for tree dependency graphs.

For each fixed ℓ , put $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$. There are a probability law ν_{δ_ℓ} supported on $\mathcal{N}$ and a finite D_ℓ such that

$$\nu_{\delta_\ell}[w] \leq D_\ell P[w]^{\delta_\ell} \quad (47)$$

for every finite tape prefix w . Consequently the same law satisfies $\nu_{\delta_\ell}[w] \leq D_\ell P[w]^s$ for every $0 \leq s \leq \delta_\ell$. Hence the global cylinder threshold (30) is sharp, including at the critical power. This statement does not assert a matching one-step conditional min-entropy law.

The clique and tree lower families are extremal. The clique has partially overlapping scopes and gives the exact one-step threshold for its graph; the trees make the maximum-degree dimension and global source threshold sharp. For general degree- d repairs with $d+1 \leq 2^{k-1}$, the present one-step lower and upper thresholds are $\log_2(d+1)$ and $\log_2 c_d$ bits. Because both lower families are extremal, a matching lower bound for nonextremal overlapping repairs remains open.

These are robustness results for the original local repair map, not stronger satisfiability criteria. The trace form gives the sharp graph-specific upper bound, while Theorem 3.3 trades at most $\log_2(5/4)$ additional bits relative to the worst-case max-degree bound for an explicit prefactor and the end-to-end randomness estimate. Proposition 3.6 also shows why logarithmic fresh-block randomness cannot be removed in that model. Randomness reuse and derandomization can have different guarantees. Viewing the same trace bound across source powers also yields the dimension and pointwise-localization statements. For uniform fixed-arity redraws this is equivalent to rescaling the $s = 1$ history count. The next application shows that this reduction to power one fails for nonuniform state-dependent repair.

3.1 Persistent local recoloring

The matrix theorem also applies when a repaired variable follows a Markov kernel from its current value rather than being reset from a fixed law. This gives a local calculation for a class in which $K_{A,s}$ is generally not a scalar multiple of $K_{A,1}$.

Proposition 3.8 (State-dependent variable repair). *Let $\Omega = \prod_{i=1}^n \mathcal{X}_i$, where the alphabets are finite. A flaw A depends on coordinates $S_A \subseteq [n]$. When A is addressed at $x \in A$, coordinate $i \in S_A$ is*

updated independently according to a stochastic matrix $P_{A,i}(x_i, \cdot)$; all other coordinates remain fixed. Then

$$K_{A,s}(x, y) = \mathbf{1}\{x_{-S_A} = y_{-S_A}, x_{S_A} \in A\} \prod_{i \in S_A} P_{A,i}(x_i, y_i)^s. \quad (48)$$

Powered repair matrices with disjoint scopes commute for every $s > 0$.

For positive vectors $z_{i,s}$ on $\mathcal{X}_i$, take v_s proportional to $\prod_i z_{i,s}$. The charge in (9) is then

$$\lambda_A(s) = \max_{y_{S_A}} \frac{\sum_{x_{S_A} \in A} \prod_{i \in S_A} z_{i,s}(x_i) P_{A,i}(x_i, y_i)^s}{\prod_{i \in S_A} z_{i,s}(y_i)}. \quad (49)$$

Thus scopes of size at most k over alphabets of size at most q have locally computable charges, in $q^{O(k)}$ arithmetic operations, without forming the q^n -state assignment chain.

Here is a concrete specialization. Orient the edges of a graph $G = (V, E)$. A coloring $x : V \rightarrow [q]$ is flawed at $e = (u, v)$ when $x_u = x_v$. Fix a positive circulant stochastic matrix Q on $[q]$. If a flawed edge has common color c , repair it by drawing a uniformly from $[q]$, drawing b independently from $Q(c, \cdot)$, and setting $(x_u, x_v) = (a, b)$. Put $D = L(G)$ and

$$R_Q(s) = \sum_{d=1}^q Q(c, d)^s, \quad \lambda_Q(s) = q^{-s} R_Q(s), \quad (50)$$

which do not depend on c by circulancy.

Theorem 3.9 (Persistent recoloring). *Start from a fixed coloring and use any deterministic nonanticipating selector. If $N_T(D)$ is the number of length- T traces of D , then*

$$Z_T^\Lambda(s) \leq q^{|V|} N_T(D) \lambda_Q(s)^T, \quad \limsup_{T \rightarrow \infty} Z_T^\Lambda(s)^{1/T} \leq \kappa_D \lambda_Q(s). \quad (51)$$

If a possibly history-dependent tape law ν satisfies $\nu[w] \leq D_0 P[w]^s$ on every live prefix, then

$$\limsup_{T \rightarrow \infty} \mathbb{P}_\nu\{\tau \geq T\}^{1/T} \leq \kappa_D \lambda_Q(s). \quad (52)$$

If $q > \kappa_D$, there is a unique $\delta_Q(D) \in (0, 1)$ satisfying

$$\kappa_D \lambda_Q(\delta_Q(D)) = 1, \quad (53)$$

and

$$\dim_H^P \mathcal{N} \leq \delta_Q(D). \quad (54)$$

Under computability, the same bound holds for the effective strong dimension of every nonterminating tape.

This family separates running time from tape geometry exactly. Let G be a matching of $m \geq 2$ edges and start with every edge flawed.

Proposition 3.10 (The same running time, different tape dimensions). *For every deterministic nonanticipating selector,*

$$\mathbb{P}\{\tau = t\} = \binom{t-1}{m-1} \left(1 - \frac{1}{q}\right)^m q^{-(t-m)}, \quad t \geq m. \quad (55)$$

Hence the complete running-time law is independent of Q and of the selector, and its exponential survival rate is $1/q$. If δ_Q is the unique solution of $\lambda_Q(\delta_Q) = 1$, then

$$\dim_H^P \mathcal{N} = \delta_Q. \quad (56)$$

For a computable process, δ_Q is also the maximal effective strong dimension of a nonterminating tape.

In particular, for

$$Q_\varepsilon = (1 - \varepsilon)I + \varepsilon J/q, \quad 0 < \varepsilon \leq 1, \quad (57)$$

all values of ε give the same law (55), whereas

$$\lambda_\varepsilon(s) = q^{-s} \left[\left(1 - \varepsilon + \frac{\varepsilon}{q}\right)^s + (q-1) \left(\frac{\varepsilon}{q}\right)^s \right], \quad \delta_1 = \frac{1}{2}, \quad \delta_\varepsilon \rightarrow 0 \quad (\varepsilon \downarrow 0). \quad (58)$$

For $0 < \varepsilon < 1$ and $s \neq 1$, the powered repair matrix is not a scalar multiple of its power-one counterpart.

The matching makes commutation nonvacuous but has disjoint repairs. For an overlapping example, take $G = P_4$ and $q = 4$. Its dependency graph is P_3 and $\kappa_{P_3} = (3 + \sqrt{5})/2$. Thus every selector and every Q_ε above satisfy

$$\limsup_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} \leq \frac{3 + \sqrt{5}}{8} < 1, \quad (59)$$

while the dimension bound is the unique root of

$$\frac{3 + \sqrt{5}}{2} 4^{-\delta} \left[\left(1 - \frac{3\varepsilon}{4}\right)^\delta + 3 \left(\frac{\varepsilon}{4}\right)^\delta \right] = 1. \quad (60)$$

It equals $(1 + \log_4((3 + \sqrt{5})/2))/2$ at $\varepsilon = 1$ and tends to $\log_4((3 + \sqrt{5})/2)$ as $\varepsilon \downarrow 0$. This overlapping example is nonextremal; no exactness claim is made for (60). The proofs are in Appendix B.

Corollary 3.11 (Rainbow perfect matchings). *Let $n \geq 2$. Color the edges of K_{2n} so that each color occurs on at most $q \geq 2$ edges, and run Kolmogorov's perfect-matching repair oracle from an arbitrary perfect matching. A flaw is a pair of vertex-disjoint edges of the same color. Put*

$$A = (2n - 1)(2n - 3), \quad C = (2n - 1)(q - 1).$$

Suppose, for some $0 \leq h \leq \log_2 A$, every repair choice has conditional atom probability at most 2^{-h} at every feasible positive-probability history. If, for some $\varepsilon > 0$,

$$h \geq \log_2 \left(\frac{256}{27} C \right) + \varepsilon,$$

then every deterministic nonanticipating selector satisfies

$$\mathbb{P}\{\tau \geq T\} \leq (2n - 1)!! e^{n/6} 2^{-\varepsilon T}. \quad (61)$$

For uniform repair choices this is nonvacuous when $q - 1 < 27(2n - 3)/256$. Writing P for the uniform oracle-action law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2((256/27)(2n - 1)(q - 1))}{\log_2((2n - 1)(2n - 3))} \right\}. \quad (62)$$

The same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$ under the computability assumptions of Corollary 2.4.

The matching oracle is a non-product-space example: its action count, atomicity, and strong commutativity are established by Kolmogorov (2018). The dimensions in (62) refer to its uniform action-outcome tree. The derivations of both applications are given in Appendix B.

4 A backward identity for individual runs

The source-power theorem sums over all live prefixes. A complementary identity controls each realized run without a commutativity or local-lemma assumption.

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. Each bad event A depends on coordinates $v(A)$ and has probability $p_A > 0$. Draw $S_0 \sim \pi$. Whenever a bad event is present, a deterministic rule selects A_t , an independent block $Z_t \sim \pi_{v(A_t)}$ is written on its coordinates, and the resulting assignment is S_t . On the event $E_T = \{\tau \geq T\}$, put

$$\Xi_T = (S_0, Z_1, \dots, Z_T), \quad P_{\text{tape}}(\Xi_T) = \pi(S_0) \prod_{t=1}^T \pi_{v(A_t)}(Z_t).$$

Appendix B gives the formal prefix tree and its source-relative metric.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the overwritten configuration and let $L_T = (A_1, \dots, A_T)$. Because A_t is true immediately before repair,

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t \mid A_t). \quad (63)$$

For any subprobability law Q_T that is positive on feasible length- T repair histories, define the backward law

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t \mid A_t). \quad (64)$$

The map

$$\Xi_T \mapsto (S_T, L_T, Y_{1:T})$$

is injective: reverse the repairs, reading each used block from the current assignment and then restoring the overwritten values.

Theorem 4.1 (Backward likelihood identity). *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (65)$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (66)$$

If the instance and $(Q_T)_{T \geq 1}$ are uniformly computable, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T \mid T, \text{instance}) \geq \Delta_T - O(1). \quad (67)$$

The likelihood ratio also gives an exact slack identity. Let $\mathcal{W}_T$ be the prefix-free set of minimal prefixes producing T repairs, let $p_T = P_{\text{tape}}(E_T)$, and put

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)), \quad \mu_T(w) = G_T(R_T(w)), \quad m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w).$$

When $p_T m_T > 0$, define $P_T(w) = P_{\text{tape}}(w)/p_T$ and $\nu_T(w) = \mu_T(w)/m_T$.

Corollary 4.2 (Exact slack and the optimal history law). *For every fixed Q_T ,*

$$-\log_2 p_T = \mathbb{E}_{P_T} \Delta_T + D_2(P_T \| \nu_T) - \log_2 m_T. \quad (68)$$

If $J_T = \sum_{t=1}^T \log_2(1/p_{A_t})$, then

$$\sup_{Q_T} \mathbb{E}[\Delta_T \mid E_T] = \mathbb{E}[J_T \mid E_T] - H_2(L_T \mid E_T), \quad (69)$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by $Q_T(\ell) = \mathbb{P}\{L_T = \ell \mid E_T\}$ on the conditional support.

Reverse reconstruction and its Kolmogorov-complexity use are established in entropy compression (Moser, 2009; Messner and Thierauf, 2012). The point of Theorem 4.1 is the common nonuniform likelihood: one ratio gives the tail, the tape-by-tape coding saving, and the exact relative-entropy slack. Proofs are in Appendix B.

5 Conclusion

The survival probability records how often a randomized computation runs for a long time. It does not determine how much randomness remains in the tapes on which the computation never stops. The source-power sum $Z_T(s)$ places these questions in one family: power one gives survival, while variation in s controls the effective and Hausdorff dimensions of nontermination.

The main theorem makes this family local. Whenever powered matrices for nonadjacent repairs commute, a full-prefix argument bounds $Z_T(s)$ by a Cartier–Foata/Shearer trace series, uniformly over history-dependent deterministic selectors. This yields weak-source termination and dimension bounds without constructing the full assignment chain. For k -SAT the graph-specific threshold is $\log_2 \kappa_D$ bits per redraw; tree and clique formulas establish the stated forms of sharpness, and recurrent clauses give an exact pointwise localization.

Persistent recoloring shows why the entire power family matters. Its nonuniform state-dependent kernels are not scalar rescalings of the ordinary transition matrices. On a matching, every persistence parameter gives the same complete running-time law, yet the nontermination dimension ranges from $1/2$ to arbitrarily near zero. Thus even complete distributional knowledge of the stopping time need not determine the geometry or individual complexity of the exceptional tapes. The backward likelihood identity gives the complementary pathwise statement: one nonuniform likelihood ratio simultaneously controls the tail, tape-by-tape compression, and its exact relative-entropy slack.

Two boundaries remain. The tree construction is sharp for dimension and global source domination, not for the one-step weak-source threshold of general nonextremal overlapping repair. More broadly, the full assignment always gives an exact finite-state operator but is usually exponential. A central open problem is to identify a natural causal or boundary width that yields a compact exact source-power representation, or to prove matching hardness for succinct bounded-scope repair.

A Finite-state and tape foundations

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0 . Each outgoing edge $e : s \rightarrow t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \dots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0 . Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (70)$$

For any square matrix C , its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ -weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

Use $P[w]$ as the diameter of the cylinder $[w]$. The resulting Hausdorff dimension is the source-relative dimension $\dim_H^P$ defined in the introduction; for a uniform q -ary source it is ordinary Hausdorff dimension in the q -adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\text{eff}, P}$ by replacing the limsup with a liminf.

Proposition A.1 (Detailed finite-state formula). *There is a unique $\delta \in [0, 1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (71)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (72)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (73)$$

If $B_\delta h = h$ with $h_s > 0$ for every state s , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (74)$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^n Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (75)$$

For a computable system with computable Q_δ , every nonterminating tape satisfies $\text{Dim}_{\text{eff},P}(\omega) \leq \delta$. A tape that is Martin–Löf random under Q_δ satisfies $\text{dim}_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$.

The requirement that Q_δ be computable is the effectivity condition needed for the pointwise conclusion. In the strict-interior finite cases below it follows automatically: if the edge probabilities are computable and

$$\rho(B_0) > 1 > \rho(B_1),$$

then $\theta \mapsto \rho(B_\theta)$ is uniformly computable and strictly decreasing, so bisection computes δ . Irreducibility makes the Perron eigenvalue simple; solving $B_\delta h = h$ with $\sum_s h_s = 1$ gives a computable positive eigenvector and hence a computable Q_δ . Endpoint roots used below are explicit.

Proof of Proposition A.1. Since the graph contains a directed cycle, $\rho(B_0) \geq 1$. Since B_1 is substochastic, $\rho(B_1) \leq 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ , because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ . The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (71); setting $\theta = 1$ gives (72).

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ -weight of the length- n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\text{dim}_H^P \mathcal{N} \leq \delta$. At $\theta = \delta$, (74) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w , such that $Q_\delta[w] \leq CP[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\text{dim}_H^P \mathcal{N} \geq \delta$. It therefore gives the required lower bound, proving (73) and (75) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system with computable Q_δ , coding under Q_δ gives, for every surviving prefix w of length n ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ -Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). $\square$

Under a uniform q -ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t . Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\text{dim}_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\text{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \quad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (75) is the finite-state form of the comparison used in Theorem 4.1.

B Proofs of the main results

The consumed-tape model. The tape is the rooted tree of source choices actually queried by the computation. A finite node is a feasible transcript $w = (s_0, z_1, \dots, z_t)$. The root branches to s_0 with probability $\pi(s_0)$. If w is live and the rule next selects A_{t+1} , its children are indexed by local blocks z , with edge probability $\pi_{v(A_{t+1})}(z)$. Hence

$$P_{\text{tape}}[w] = \pi(s_0) \prod_{j=1}^t \pi_{v(A_j)}(z_j). \quad (76)$$

Terminal transcripts are leaves, and infinite branches are exactly the nonterminating consumed tapes. For two distinct infinite branches, let $\omega \wedge \eta$ be their longest common consumed prefix and set

$$d_P(\omega, \eta) = P_{\text{tape}}[\omega \wedge \eta]. \quad (77)$$

All dimension statements assume that these cylinder masses tend to zero along every infinite branch.

Fixed computable enumerations of the outcome alphabets give a canonical self-delimiting encoding of each transcript. The dimensions are unchanged under a computable prefix-tree isomorphism Ψ with computable inverse and bounded source distortion

$$C^{-1} P_{\text{tape}}[w] \leq P'_{\text{tape}}[\Psi(w)] \leq C P_{\text{tape}}[w],$$

because self-information and prefix complexity then change by at most a bounded additive term. No invariance under arbitrary recodings is asserted. A pre-sampled resampling table is only a coupling device: unused entries are excluded, since adjoining independent unused coordinates can change tape dimension without changing the computation.

Backward likelihood.

Proof of Theorem 4.1. Take the ratio of (64) and (63). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (65).

It remains to use this likelihood ratio. Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce at least T repairs, and let $\mathcal{R}_T$ be the set of formal records on which G_T is defined. For $w \in \mathcal{W}_T$, write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that R_T is injective.

First observe that G_T has total mass at most one. Indeed, for a fixed list $\ell = (A_1, \dots, A_T)$, the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of R_T gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus μ_T is a subprobability mass function. Its total mass can be strictly less than one because Q_T may itself have mass less than one, or because G_T may assign mass to formal records that are not produced by the algorithm.

For every $w \in \mathcal{W}_T$, the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because $\Delta_T = -\infty$ outside E_T and the cylinders determined by $\mathcal{W}_T$ are disjoint, for every $u \geq 0$,

$$\begin{aligned} \mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\ &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\ &\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\ &\leq 2^{-u}. \end{aligned}$$

This proves (66); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Use the canonical effective encoding of the consumed-choice tree from Section B, and extend μ_T by zero off $\mathcal{W}_T$. Membership in $\mathcal{W}_T$ is decidable uniformly from (T, w) by simulating the deterministic computation, and $w \mapsto R_T(w)$ is uniformly computable. The finite source factors and Q_T are uniformly computable by hypothesis. Hence $(\mu_T)_{T \geq 1}$ is a uniformly lower semicomputable family of subprobability masses (indeed, uniformly computable). The Kraft–Chaitin coding theorem for lower semicomputable semimeasures, equivalently Shannon–Fano coding in this discrete setting (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019), gives

$$K(\Xi_T \mid T, \text{instance}) \leq -\log_2 \mu_T(\Xi_T) + O(1).$$

The constant is uniform in T and in the realized run. Since

$$-\log_2 \mu_T(\Xi_T) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T,$$

rearranging proves (67). □

Proof of Corollary 4.2. On $\mathcal{W}_T$,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking P_T -expectations gives (68). For the second claim, $\Delta_T = J_T + \log_2 Q_T(L_T)$. If r_T is the conditional law of L_T given E_T , then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \| Q_T)$$

when Q_T has total mass one; allocating less mass can only decrease the left side. The maximum is attained at $Q_T = r_T$. $\square$

Powered matrix theorem.

Proof of Lemma 2.1. The (σ, τ) entry of $K_{f,s} K_{g,s}$ is the sum over all action-labelled two-step paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

of

$$(\rho_f(a \mid \sigma) \rho_g(b \mid \sigma'))^s.$$

The diamond bijection maps these paths onto the corresponding g, f paths with the same endpoints and preserves each product before it is powered. The two matrix entries are therefore equal.

For the fixed-arity statement, let $m_f(\sigma, \tau)$ be the number of action labels that move σ to τ when f is addressed. If f is present at σ , then

$$K_{f,s}(\sigma, \tau) = m_f(\sigma, \tau) M_f^{-s} = M_f^{1-s} K_{f,1}(\sigma, \tau).$$

Both sides vanish if f is absent. Hence, for nonadjacent f, g whose ordinary matrices commute,

$$K_{f,s} K_{g,s} = M_f^{1-s} M_g^{1-s} K_{f,1} K_{g,1} = K_{g,s} K_{f,s}.$$

$\square$

Proof of Lemma 2.2. First observe that $K_{H,s}$ is well defined. If x and y are two sources of H , their labels are nonadjacent; otherwise H would contain an edge between them. Their powered matrices commute, and induction on $|H|$ shows that deleting the sources in either order gives the same product.

A stronger conditional statement is proved by induction. Fix an arbitrary past history $\mathfrak{h}$ ending at state σ , and let $M_{\mathfrak{h}}(H)$ be the powered mass of all continuations whose next $|H|$ repairs induce H . The claim is

$$M_{\mathfrak{h}}(H) \leq e_{\sigma}^{\top} K_{H,s} \mathbf{1}. \quad (78)$$

It is immediate for $H = \emptyset$.

Suppose $H \neq \emptyset$, and let f be the flaw selected after $\mathfrak{h}$. If no source of H is labelled f , then $M_{\mathfrak{h}}(H) = 0$. Otherwise there is a unique such source x , since two vertices with the same label are adjacent and cannot both be sources. After an action $a \in \mathcal{A}_f(\sigma)$ leading to $\tau = \Phi_f(\sigma, a)$, the

remaining repairs must induce $H - x$. Applying the induction hypothesis separately after every possible action gives

$$\begin{aligned} M_{\mathfrak{h}}(H) &\leq \sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma)^s e_{\Phi_f(\sigma, a)}^\top K_{H-x, s} \mathbf{1} \\ &= e_\sigma^\top K_{f, s} K_{H-x, s} \mathbf{1} = e_\sigma^\top K_{H, s} \mathbf{1}. \end{aligned}$$

Actions leading to the same state are added in $K_{f, s}$; no recovery of an action from its endpoints is used. The induction is uniform over $\mathfrak{h}$, so the selector may use the complete past action history.

Finally, sum (78) over the initial source symbols. Their powered masses at state σ sum to $b_s(\sigma)$, which gives (14). $\square$

Proof of Theorem 2.3. For every full history DAG H , the definition of $\lambda_f(s)$ gives

$$v_s^\top K_{H, s} \mathbf{1} \leq \left(\prod_{x \in H} \lambda_{L(x)}(s) \right) v_s^\top \mathbf{1} = \prod_{x \in H} \lambda_{L(x)}(s).$$

Since $b_s \leq \gamma_s v_s$ coordinatewise, Lemma 2.2 yields

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq \gamma_s \prod_{x \in H} \lambda_{L(x)}(s). \quad (79)$$

The full history DAG of a flaw word is its labelled dependence graph. Full history DAGs, understood up to the label-preserving isomorphism fixed above, are therefore in one-to-one correspondence with traces; only a subset need be realizable by the algorithm. Summing (79) over realized DAGs and then enlarging to all traces of length T proves

$$Z_T^\Lambda(s) \leq \gamma_s \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_f \lambda_f(s)^{N_f(h)} = \gamma_s a_T(s).$$

The Cartier–Foata identity gives the asserted term-by-term formal-series inequality.

If $q\lambda(s)$ lies in the open Shearer region, the series in (13) converges at $z = q$ and has nonnegative terms. Hence

$$q^T a_T(s) \leq \sum_{t \geq 0} q^t a_t(s) = \frac{1}{P_D(q\lambda(s))},$$

which proves (18).

It remains to derive the local cluster estimate. Assume first that $0 < \theta_s < 1$ and set

$$\bar{\lambda}_f = \frac{\lambda_f(s)}{\theta_s}.$$

By definition of θ_s ,

$$\bar{\lambda}_f \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I) \leq \eta_f.$$

The restricted cluster-expansion enumeration therefore gives, for every independent sink set R ,

$$\sum_{\substack{h \in \mathcal{M}(D) \\ \text{sink}(h)=R}} \prod_f \bar{\lambda}_f^{N_f(h)} \leq \eta(R). \quad (80)$$

This is the standard trace-DAG form of the cluster bound; see Harris et al. (2025, Proposition 3.9). Restricting the sum to traces of length T and restoring the factor θ_s^T gives

$$a_T(s) \leq \theta_s^T \sum_{R \in \text{Ind}(F)} \eta(R).$$

Together with (16), this proves (19). If $\theta_s = 0$, then every $\lambda_f(s) = 0$, so the assertion is immediate for $T \geq 1$; at $T = 0$ it follows from $\sum_{\sigma} b_s(\sigma) \leq \gamma_s$. $\square$

Proof of Corollary 2.4. The depth- T cylinders indexed by $\mathcal{L}_T^\Lambda$ cover $\mathcal{N}$ and have source diameters at most a fixed initial factor times $\rho_{\max}^T$. By (20),

$$\sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s \leq C_s \vartheta_s^T \longrightarrow 0.$$

This proves that the source-relative s -dimensional Hausdorff measure of $\mathcal{N}$ is zero.

For the effective statement, define on $\mathcal{L}_T^\Lambda$

$$\kappa_{T,s}(w) = \frac{P_{\text{tape}}[w]^s}{\widehat{C}_s \widehat{\vartheta}_s^T}.$$

These are subprobability masses because

$$\sum_{w \in \mathcal{L}_T^\Lambda} \kappa_{T,s}(w) \leq \frac{C_s \vartheta_s^T}{\widehat{C}_s \widehat{\vartheta}_s^T} \leq 1.$$

They are uniformly computable under the stated assumptions. Prefix-free coding gives

$$\begin{aligned} K(w \mid T, \text{instance}) &\leq -\log_2 \kappa_{T,s}(w) + O(1) \\ &= s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1). \end{aligned}$$

Removing the condition T costs $O(\log T)$. Since $\rho_{\max} < 1$, the source self-information of a depth- T prefix grows at least linearly in T . Divide by this self-information and take a limsup; the term $T \log_2 \widehat{\vartheta}_s$ is nonpositive, and the result follows. $\square$

Proof of Corollary 2.5. The cylinders indexed by $\mathcal{L}_T^\Lambda$ are disjoint. Hence

$$\mathbb{P}_\nu\{\tau \geq T\} = \sum_{w \in \mathcal{L}_T^\Lambda} \nu[w] \leq D \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s.$$

Apply (20). $\square$

Weak-randomness applications.

Proof of Proposition 3.8. The matrix in (48) acts only on the tensor factors indexed by S_A . If $S_A \cap S_B = \emptyset$, neither repair changes whether the other flaw is present, and the two tensor operators commute.

Let $Z_s = \prod_i \sum_a z_{i,s}(a)$. For a terminal state y ,

$$\sum_x v_s(x) K_{A,s}(x, y) = \frac{\prod_{j \notin S_A} z_{j,s}(y_j)}{Z_s} \sum_{x_{S_A} \in A} \prod_{i \in S_A} z_{i,s}(x_i) P_{A,i}(x_i, y_i)^s.$$

Dividing by $v_s(y) = Z_s^{-1} \prod_i z_{i,s}(y_i)$ gives (49). Changing S_A can create only flaws whose scopes meet S_A , so scope intersection is a valid potential-causality graph. The final complexity claim follows by direct enumeration of the local states. $\square$

Proof of Theorem 3.9. Repairs of nonincident edges act on disjoint vertex pairs, so Proposition 3.8 gives powered commutation for every $s > 0$. Take v_s uniform on $[q]^V$. For an oriented edge $e = (u, v)$ and fixed terminal colors (a, b) , the local column sum over flawed prestates is

$$\sum_{c=1}^q (q^{-1} Q(c, b))^s = q^{-s} R_Q(s) = \lambda_Q(s).$$

Circulancy makes the sum independent of b . A fixed initial coloring has $\gamma_s = q^{|V|}$. Since all charges are equal, Theorem 2.3 gives

$$Z_T^\Lambda(s) \leq q^{|V|} N_T(D) \lambda_Q(s)^T,$$

which proves (51). Combining this estimate with cylinder domination proves (52).

Every action atom $q^{-1} Q(c, b)$ is strictly below one. Moreover, $\lambda_Q(0) = q$ and, because a circulant stochastic matrix is doubly stochastic, $\lambda_Q(1) = 1/q$. Positivity of Q makes $\lambda_Q(s)$ continuous and strictly decreasing. If $q > \kappa_D$, the critical equation therefore has a unique solution in $(0, 1)$. For every $s > \delta_Q(D)$, the right side of (51) decays exponentially. Apply Corollary 2.4 and let $s \downarrow \delta_Q(D)$. $\square$

Proof of Proposition 3.10. At each repair, conditional on the complete past and on the common color c , the selected edge remains flawed with probability

$$\sum_{d=1}^q q^{-1} Q(c, d) = \frac{1}{q}.$$

A successful repair permanently removes that edge because the edges are disjoint. The success indicators are therefore Bernoulli trials with conditional success probability $1 - 1/q$, and termination occurs at the m th success. This gives (55).

The line graph of a matching is edgeless, so its trace growth is one. Theorem 3.9 gives the upper dimension bound by taking $s > \delta_Q$ and then $s \downarrow \delta_Q$. For the reverse bound, at the critical power define, whenever the selected edge has common color c ,

$$\nu_{\delta_Q}\{(a, b) = (d, d) \mid c\} = (q^{-1} Q(c, d))^{\delta_Q}. \quad (81)$$

These probabilities sum to one by $\lambda_Q(\delta_Q) = 1$. They keep every selected edge flawed; since all edges start flawed, the resulting law is supported on $\mathcal{N}$. On every prefix in its support,

$$\nu_{\delta_Q}[w] = P[w]^{\delta_Q}.$$

The mass-distribution principle yields $\dim_H^P \mathcal{N} \geq \delta_Q$.

Under computability, choose a Martin–Löf-random tape for the computable Markov law (81). The Levin–Schnorr theorem gives $K(w) \geq -\log_2 \nu_{\delta_Q}[w] - O(1)$ along its prefixes, while coding for computable measures gives $K(w) \leq -\log_2 \nu_{\delta_Q}[w] + O(\log |w|)$. Every reference action atom is at

most $1/q$, so $I_P(w)$ grows at least linearly in $|w|$. Since $-\log_2 \nu_{\delta_Q}[w] = \delta_Q I_P(w)$, the resulting tape has effective strong P -dimension δ_Q ; the reverse inequality follows from the upper bound.

For Q_ε , direct substitution gives (58). At $\varepsilon = 1$, $\lambda_1(s) = q^{1-2s}$ and hence $\delta_1 = 1/2$. For each fixed $s > 0$, $\lambda_\varepsilon(s) \rightarrow q^{-s} < 1$ as $\varepsilon \downarrow 0$, which forces $\delta_\varepsilon \rightarrow 0$. Finally, for $0 < \varepsilon < 1$ the diagonal and off-diagonal entries of Q_ε have different ratios after taking a power $s \neq 1$, so no common scalar can turn $K_{e,1}$ into $K_{e,s}$. $\square$

For $G = P_4$, the line graph is P_3 and

$$P_{P_3}(z\mathbf{1}) = 1 - 3z + z^2, \quad \kappa_{P_3} = \frac{3 + \sqrt{5}}{2}.$$

Equations (59) and (60) now follow from Theorem 3.9 and (58). Substitution of $\varepsilon = 1$ and passage to the limit $\varepsilon \downarrow 0$ give the two stated roots.

For the variable-model calculation, take

$$v_s(\sigma) = \frac{\pi(\sigma)^s}{\sum_{\xi \in \Omega} \pi(\xi)^s}.$$

For $S = v(A)$ and a fixed terminal assignment τ ,

$$\begin{aligned} \sum_{\sigma} v_s(\sigma) K_{A,s}(\sigma, \tau) &= v_s(\tau) \sum_{b \in A} \pi_s(b)^s \\ &= v_s(\tau) r_A(s). \end{aligned}$$

This proves (26).

Proof of Theorem 3.1. For uniform k -clause resampling, $\lambda_A(s) = 2^{-ks}$ for every clause A . A fixed initial assignment and the uniform choice of v_s give $\gamma_s = 2^n$. Therefore Theorem 2.3 gives

$$Z_T^\Lambda(s) \leq 2^n 2^{-ksT} |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

For every $\xi > \kappa_D$, the Cartier–Foata series at $z = \xi^{-1}$ gives

$$|\{h \in \mathcal{M}(D) : |h| = T\}| \leq \frac{\xi^T}{P_D(\xi^{-1}\mathbf{1})}.$$

Multiplying by D_0 proves (29). A conditional atom bound implies $\nu[w] \leq 2^{-hT} = P[w]^{h/k}$ by the chain rule, so this is the special case $s = h/k$ and $D_0 = 1$.

For every $s > \log_2(\kappa_D)/k$, choose $\xi \in (\kappa_D, 2^{ks})$. The resulting source-power sum decays exponentially, which proves the Hausdorff and, under computability, effective dimension bounds by Corollary 2.4. Finally, the symmetric form of Shearer’s theorem (Shearer, 1985; Scott and Sokal, 2005) states that a graph of maximum degree $d \geq 2$ has $z\mathbf{1}$ in its open Shearer region whenever

$$0 \leq z < \frac{(d-1)^{d-1}}{d^d}.$$

Hence $R_D \geq (d-1)^{d-1}/d^d$ and $\kappa_D \leq c_d$. $\square$

Proof of Corollary 3.2. Fix a nonempty $R \subseteq F$. If $\text{Rec}(\omega) = R$, every clause outside R is repaired only finitely often. After a finite tape prefix, the run therefore repairs only clauses in R . Fix such a prefix u . On a history of the instance that retains only the clauses in R , prepend u and apply the original selector whenever it chooses a currently violated clause in R ; otherwise choose the first currently violated clause of R in a fixed order. This defines a deterministic nonanticipating selector for the reduced instance and agrees with the original run on every continuation after u whose later repair labels all lie in R . It is computable whenever the original selector is. The proof of Theorem 3.1 now applies to the induced graph $D[R]$. For every $s > \log_2(\kappa_{D[R]})/k$, the source-power sum of the suffix decays exponentially.

There are countably many finite prefixes after which the run remains in R . Prefixing rescales the source metric by a constant, and Hausdorff dimension is stable under countable unions. This proves the setwise bound. In the computable case, a fixed prefix and the finite set R cost only an additive constant in a program description. Prefix-free coding of the same suffix bound and passage to the infimum over s prove (34). $\square$

Proof of Theorem 3.3. For a violated k -clause under the uniform reference source, precisely one local assignment is bad, so its powered charge is 2^{-ks} . Take v_s uniform on the 2^n assignments. A fixed initial assignment gives $\gamma_s = 2^n$. Set $\eta_A = \eta = 1/(d-1)$. A clause is adjacent to every other member of its inclusive neighborhood. An independent subset of that neighborhood therefore either consists of the clause itself or is contained in its at most d neighbors. Hence

$$\sum_{I \in \text{Ind}(\Gamma(A))} \eta(I) \leq \eta + (1 + \eta)^d.$$

Consequently

$$\theta_s \leq 2^{-ks} \frac{\eta + (1 + \eta)^d}{\eta} = 2^{-ks} \left(1 + \frac{d^d}{(d-1)^{d-1}} \right) = 2^{-ks} A_d,$$

whereas

$$\sum_{R \in \text{Ind}(F)} \eta(R) \leq (1 + \eta)^m = \left(\frac{d}{d-1} \right)^m.$$

The atom condition implies that every length- T redraw word has probability at most 2^{-hT} . With $s = h/k$, this equals its uniform reference probability raised to s . If $h = \log_2 A_d + \varepsilon$, apply Corollary 2.5 to obtain (37).

Taking $h = k$ gives (38). For the dimension statement, every $s > \log_2 A_d/k$ makes $\theta_s < 1$. Corollary 2.4 applies for every such s ; let s decrease to the displayed value. The bound by one is automatic. $\square$

Proof of Corollary 3.4. Conditioned on the past, injectivity of E_t makes every supported redraw have probability exactly 2^{-r} . The encoder is chosen before U_t , so its dependence on the past does not change this fact. Apply Theorem 3.3 with $h = r$. For a bound $\mathbb{P}\{\tau \geq T\} \leq C_0 2^{-\varepsilon T}$, bound the first $\lceil \log_2(C_0)/\varepsilon \rceil$ terms of $\mathbb{E}\tau = \sum_{T \geq 1} \mathbb{P}\{\tau \geq T\}$ by one and sum the remaining geometric series. With $C_0 = 2^n(d/(d-1))^m$ this gives (40); every repair consumes exactly r bits.

If $A_d \leq 2^{k-1}$ and $r = \lceil \log_2 A_d + 1 \rceil$, then $r \leq k$, $\varepsilon_r \geq 1$, and $r = O(\log(d+1))$. Moreover, $\log_2(d/(d-1)) = O(1/d)$ for $d \geq 2$. Substitution in (40) proves (41). $\square$

Proof of Corollary 3.5. Condition on the entire past before a block is read. The chain rule and the Santha–Vazirani inequalities give, for every $z \in \{0, 1\}^r$,

$$\mathbb{P}\{(U_{j+1}, \dots, U_{j+r}) = z \mid U_1, \dots, U_j\} \leq (1 - \alpha)^r.$$

After an injective map, the same upper bound holds for every redraw atom. Thus Theorem 3.3 applies with $h = r \log_2(1/(1 - \alpha))$. $\square$

Proof of Proposition 3.6. Put $\ell = \lceil \log_2(d + 1) \rceil$ and choose distinct words $u_1, \dots, u_{d+1} \in \{0, 1\}^\ell$. Use ℓ common variables and, for each i , a disjoint private block P_i of $k - \ell$ variables. Let C_i be the unique clause on the common variables and P_i that is false when the common word is u_i and P_i is all zero. The clauses share the common variables, so their dependency graph is K_{d+1} , but their scopes are not equal. Distinct clauses cannot be false simultaneously. The formula is satisfiable, for example by setting one private variable in each P_i to one.

Start with every private block zero and the common word equal to u_1 . Call i active when P_i is still zero and let q be the number of active indices. In every live state the common word is u_i for a unique active i , so C_i is the only violated clause. Write $M = 2^{k-\ell}$. When C_i is repaired, exactly q of its 2^k redraws leave q unchanged: the new private block is zero and the new common word is u_j for an active j . Exactly $(q - 1)(M - 1)$ redraws move to $q - 1$: the new private block is nonzero and the new common word is u_j for one of the other active indices. Every other redraw terminates. Thus the all-power live matrix on $q = 1, \dots, d + 1$ is bidiagonal, with

$$R_s(q, q) = q 2^{-ks}, \quad R_s(q, q - 1) = (q - 1)(M - 1) 2^{-ks}. \quad (82)$$

Its spectral radius is $(d + 1) 2^{-ks}$. Proposition A.1 therefore gives (42) and (43), including pointwise attainment.

At $q = d + 1$, instead redraw the selected scope uniformly from the $d + 1$ assignments $(u_j, 0)$, $1 \leq j \leq d + 1$. This source has conditional block min-entropy $\log_2(d + 1)$ and remains at $q = d + 1$ forever. If $d + 1 = 2^r$, the redraw is the image of an injective map applied to r fresh fair bits. Finally,

$$\frac{c_d}{d + 1} = \frac{d}{d + 1} \left(1 + \frac{1}{d - 1} \right)^{d-1} < e,$$

which proves the stated gap. $\square$

Proof of Theorem 3.7. Orient every edge of $T_{d,\ell}$ away from the root and introduce one Boolean variable X_e for each edge. For a vertex v , form a clause C_v containing one literal for each incident edge, using opposite signs at the two endpoints. Add $k - \deg(v)$ private variables to C_v , each used in no other clause. This is possible because $\deg(v) \leq d \leq k$. Every clause has exactly k distinct variables, and every variable occurs once or twice.

Two clauses share a variable exactly when their vertices are adjacent. Their bad events prescribe opposite values to the shared edge variable and are therefore disjoint; nonadjacent clauses have disjoint scopes. The formula is satisfiable. For example, use the sign convention in which the parent literal on an edge is X_e and the child literal is $\neg X_e$, set every $X_e = 0$, and set one private literal of the root to true. Every nonroot clause is satisfied by its parent-edge literal, while the root has a private literal because its degree is $d - 1 < k$.

Every bad event is atomic and has probability 2^{-k} . For an extremal instance, the standard probability-weighted trace identity (Guo et al., 2019; Jerrum, 2024) gives

$$\lim_{T \rightarrow \infty} Z_T(s)^{1/T} = 2^{-ks} \kappa_{T_{d,\ell}}.$$

At $s = 1$ this is (44); the unique power at which the displayed rate equals one is $\log_2(\kappa_{T_{d,\ell}})/k$, which gives (45). The finite-state formula above gives effective attainment under computability.

It remains to calculate the trace growth. Put $b = d - 1$ and define

$$r_0(z) = z, \quad r_{j+1}(z) = \frac{z}{(1 - r_j(z))^b}.$$

As long as the denominators are positive, $r_j(z)$ is the signed root-occupation ratio for a full b -ary subtree of height j . The first positive zero $z_{d,\ell}$ of $P_{T_{d,\ell}}(z\mathbf{1})$ is characterized by $r_\ell(z_{d,\ell}) = 1$. The critical value for the iteration is

$$z_* = \max_{0 \leq r < 1} r(1-r)^b = \frac{b^b}{(b+1)^{b+1}} = \frac{(d-1)^{d-1}}{d^d}.$$

If $z < z_*$, the iteration increases to the smaller fixed point of $r = z/(1-r)^b$ and remains below one. If $z > z_*$, there is no fixed point in $[0, 1)$, so the iteration reaches one after finitely many steps. Hence $z_{d,\ell} \downarrow z_*$ and

$$\kappa_{T_{d,\ell}} = z_{d,\ell}^{-1} \uparrow z_*^{-1} = c_d.$$

Finally put $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$. For the fixed selector, consider the finite live chain on full assignments. The exact growth calculation above and the finite-state decomposition give a reachable irreducible component C whose powered transition matrix satisfies $\rho(B_{\delta_\ell, C}) = 1$. Choose a positive right Perron vector h and a live prefix u entering C . On an internal edge $e : x \rightarrow y$ of C , set

$$Q(e \mid x) = P(e)^{\delta_\ell} \frac{h(y)}{h(x)}.$$

Its row sums are one, so prefixing Q by u defines a probability law ν_{δ_ℓ} supported on $\mathcal{N}$. Telescoping along extensions of u , and absorbing the source probability of u and the finitely many ratios of entries of h into D_ℓ , gives (47) for every finite prefix w . Since $P[w] \leq 1$, the same inequality holds with $P[w]^s$ for every $0 \leq s \leq \delta_\ell$. $\square$

Proof of Corollary 3.11. For the matching oracle of [Kolmogorov \(2018\)](#), every flaw has

$$A = (2n-1)(2n-3)$$

equiprobable actions, distinct oracle executions have distinct outputs in the action-expanded graph, and the oracle is strongly commutative. Thus its powered charge is A^{-s} . Put $C = (2n-1)(q-1)$. The neighborhood of a flaw is covered by four families, one for each endpoint, each containing at most C flaws. Therefore, for constant η ,

$$\sum_{I \in \text{Ind}(\Gamma(f))} \eta^{|I|} \leq (1 + C\eta)^4.$$

Taking $\eta = 1/(3C)$ gives

$$\theta_s \leq \frac{A^{-s}}{\eta} (1 + C\eta)^4 = \frac{256}{27} C A^{-s}.$$

There are at most

$$|F| \leq \frac{n(2n-1)(q-1)}{2}$$

flaws, and hence

$$\sum_{R \in \text{Ind}(F)} \eta^{|R|} \leq (1 + \eta)^{|F|} \leq e^{|F|\eta} \leq e^{n/6}.$$

With v_s uniform on the $(2n-1)!!$ perfect matchings, a fixed initial matching has $\gamma_s = (2n-1)!!$. The conditional atom bound and the uniform reference oracle give power domination with $s = h/\log_2 A$. Corollary 2.5 now proves (61). Letting s decrease to $\log_2((256/27)C)/\log_2 A$ proves the dimension bound. Under uniform actions, the strict condition $\theta_1 < 1$ is equivalent to $q-1 < 27(2n-3)/256$. $\square$

Acknowledgements

The author used ChatGPT and Codex as research and editorial aids. The author assumes full responsibility for the statements and proofs in the paper.

References

- Samy Abbes and Jean Mairesse. Uniform and Bernoulli measures on the boundary of trace monoids. *Journal of Combinatorial Theory, Series A*, 135:201–236, 2015. doi: 10.1016/j.jcta.2015.05.003.
- Krishna B. Athreya, John M. Hitchcock, Jack H. Lutz, and Elvira Mayordomo. Effective strong dimension, algorithmic information, and computational complexity. *SIAM Journal on Computing*, 37(3):671–705, 2007. doi: 10.1137/S0097539703446912.
- Peter L. Bartlett, Olivier Bousquet, and Shahar Mendelson. Local Rademacher complexities. *The Annals of Statistics*, 33(4):1497–1537, 2005. doi: 10.1214/009053605000000282.
- Patrick Billingsley. *Ergodic Theory and Information*. John Wiley & Sons, 1965.
- Rodrigo Bissacot, Roberto Fernández, Aldo Procacci, and Benedetto Scoppola. An improvement of the lovász local lemma via cluster expansion. *Combinatorics, Probability and Computing*, 20(5):709–719, 2011.
- Rufus Bowen. *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms*, volume 470 of *Lecture Notes in Mathematics*. Springer, 1975. doi: 10.1007/BFb0081029.
- Pierre Cartier and Dominique Foata. *Problèmes combinatoires de commutation et réarrangements*, volume 85 of *Lecture Notes in Mathematics*. Springer, 1969.
- Benny Chor and Oded Goldreich. Unbiased bits from sources of weak randomness and probabilistic communication complexity. *SIAM Journal on Computing*, 17(2):230–261, 1988. doi: 10.1137/0217015.
- Dean Doron, Dana Moshkovitz, Justin Oh, and David Zuckerman. Almost Chor–Goldreich sources and adversarial random walks. In *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*, pages 1–9, 2023. doi: 10.1145/3564246.3585134.
- Paul Erdős and László Lovász. Problems and results on 3-chromatic hypergraphs and some related questions. In András Hajnal, Richard Rado, and Vera T. Sós, editors, *Infinite and Finite Sets, Vol. II*, volume 10 of *Colloquia Mathematica Societatis János Bolyai*, pages 609–627. North-Holland, Amsterdam, 1975.
- Kenneth Falconer. *Fractal Geometry: Mathematical Foundations and Applications*. Wiley, third edition, 2014. doi: 10.1002/9781119942399.
- Robert M. Fano. The transmission of information. Technical Report 65, Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, Massachusetts, 1949.
- Paula M. S. Fialho, Bernardo N. B. de Lima, and Aldo Procacci. Moser–Tardos resampling algorithm, entropy compression method and the subset gas. *Annales de l’Institut Henri Poincaré D*, 9(3):435–471, 2022. doi: 10.4171/AIHPD/122.
- Heng Guo, Mark Jerrum, and Jingcheng Liu. Uniform sampling through the Lovász Local Lemma. *Journal of the ACM*, 66(3):18:1–18:31, 2019. doi: 10.1145/3310131.
- Bernhard Haeupler and David G. Harris. Parallel algorithms and concentration bounds for the Lovász Local Lemma via witness DAGs. *ACM Transactions on Algorithms*, 13(4):53:1–53:40, 2017.

- David G. Harris, Fotios Iliopoulos, and Vladimir Kolmogorov. A new notion of commutativity for the algorithmic Lovász local lemma. *Theory of Computing*, 21(5):1–34, 2025. doi: 10.4086/toc.2025.v021a005.
- Mark Jerrum. Fundamentals of partial rejection sampling. *Probability Surveys*, 21:171–199, 2024. doi: 10.1214/24-PS29.
- Gerhard Keller and Carlangelo Liverani. Rare events, escape rates and quasistationarity: Some exact formulae. *Journal of Statistical Physics*, 135(3):519–534, 2009. doi: 10.1007/s10955-009-9747-8.
- Vladimir Kolmogorov. Commutativity in the algorithmic Lovász local lemma. *SIAM Journal on Computing*, 47(6):2029–2056, 2018. doi: 10.1137/16M1093306.
- Vladimir Koltchinskii. Local Rademacher complexities and oracle inequalities in risk minimization. *The Annals of Statistics*, 34(6):2593–2656, 2006. doi: 10.1214/009053606000001019.
- Ming Li and Paul M. B. Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, Cham, 4 edition, 2019. doi: 10.1007/978-3-030-11298-1.
- Shaojie Li and Yunbei Xu. Pointwise generalization in deep neural networks. [arXiv preprint arXiv:2605.18598](#), 2026.
- Jack H. Lutz. The dimensions of individual strings and sequences. *Information and Computation*, 187(1):49–79, 2003. doi: 10.1016/S0890-5401(03)00187-0.
- Jack H. Lutz. A divergence formula for randomness and dimension. *Theoretical Computer Science*, 412(1–2):166–177, 2011. doi: 10.1016/j.tcs.2010.09.005.
- R. Daniel Mauldin and S. C. Williams. Hausdorff dimension in graph directed constructions. *Transactions of the American Mathematical Society*, 309(2):811–829, 1988. doi: 10.1090/S0002-9947-1988-0961615-4.
- Jochen Messner and Thomas Thierauf. A Kolmogorov complexity proof of the Lovász Local Lemma for satisfiability. *Theoretical Computer Science*, 461:55–64, 2012. doi: 10.1016/j.tcs.2012.06.005.
- Robin A. Moser. A constructive proof of the Lovász Local Lemma. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing*, pages 343–350, 2009. doi: 10.1145/1536414.1536462.
- Robin A. Moser and Gábor Tardos. A constructive proof of the general Lovász Local Lemma. *Journal of the ACM*, 57(2):11:1–11:15, 2010. doi: 10.1145/1667053.1667060.
- William Parry. Intrinsic Markov chains. *Transactions of the American Mathematical Society*, 112: 55–66, 1964. doi: 10.1090/S0002-9947-1964-0161372-1.
- Wesley Pegden. An extension of the Moser–Tardos algorithmic local lemma. *SIAM Journal on Discrete Mathematics*, 28(2):911–917, 2014.
- Alfréd Rényi. On measures of entropy and information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, pages 547–561. University of California Press, 1961.
- Miklos Santha and Umesh V. Vazirani. Generating quasi-random sequences from semi-random sources. *Journal of Computer and System Sciences*, 33(1):75–87, 1986. doi: 10.1016/0022-0000(86)90044-9.

- Alexander D. Scott and Alan D. Sokal. The repulsive lattice gas, the independent-set polynomial, and the Lovász Local Lemma. *Journal of Statistical Physics*, 118:1151–1261, 2005. doi: 10.1007/s10955-004-2055-4.
- Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, July 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.
- James B. Shearer. On a problem of Spencer. *Combinatorica*, 5(3):241–245, 1985. doi: 10.1007/BF02579368.
- Stephen G. Simpson. Symbolic dynamics: Entropy = dimension = complexity. *Theory of Computing Systems*, 56(3):527–543, 2015. doi: 10.1007/s00224-014-9546-8.
- Ludwig Staiger. Kolmogorov complexity and hausdorff dimension. *Information and Computation*, 103(2):159–194, 1993. doi: 10.1006/inco.1993.1017.
- Yunbei Xu. Bellman-sufficient information complexity. [arXiv preprint arXiv:2606.11171](#), 2026a.
- Yunbei Xu. Pointwise complexity for Gaussian fields: Upper envelopes, algorithmic lower bounds, and separation. [arXiv preprint arXiv:2606.07931](#), 2026b.
- Yunbei Xu and Assaf Zeevi. Towards optimal problem dependent generalization error bounds in statistical learning theory. *Mathematics of Operations Research*, 50(1):40–67, 2025. doi: 10.1287/moor.2021.0076.